

Section 3.1 Data Preprocessing

Equations, with every symbol defined

Notation

$n = 1, \dots, N$	observation (batch)
$i = 1, \dots, B$	predictor block
$j = 1, \dots, K_i$	column within block i
K_i	number of columns in block i
N_{tr}	number of observations in the training fold
$x_{nj}^{(i)}$	raw entry: row n , column j , block i
$\tilde{x}_{nj}^{(i)}$	the same entry after preprocessing

Scaling parameters, estimated on the training fold only

$$\begin{aligned}\mu_j^{(i)} &= \frac{1}{N_{tr}} \sum_{n \in \text{train}} x_{nj}^{(i)} \\ \sigma_j^{(i)} &= \sqrt{\frac{1}{N_{tr}-1} \sum_{n \in \text{train}} (x_{nj}^{(i)} - \mu_j^{(i)})^2}\end{aligned}\tag{3.1}$$

Guard on near-constant columns

$$\sigma_j^{(i)} \leftarrow 1 \quad \text{if} \quad \sigma_j^{(i)} < \varepsilon\tag{3.2}$$

ε is a numerical tolerance (10^{-12} here). Such a column is centred but not rescaled.

Autoscaling and block scaling

$$\tilde{x}_{nj}^{(i)} = \frac{x_{nj}^{(i)} - \mu_j^{(i)}}{\sigma_j^{(i)} \sqrt{K_i}}\tag{3.3}$$

The first factor equalises the variance of the columns; the second equalises the weight of the blocks.

Applied with the same $\mu_j^{(i)}$ and $\sigma_j^{(i)}$ to the training AND the held-out observations.

Response scaling and back-transformation

$$\tilde{y}_n = \frac{y_n - \mu_y}{\sigma_y} \qquad \hat{y}_n = \sigma_y \hat{u}_n + \mu_y\tag{3.4}$$